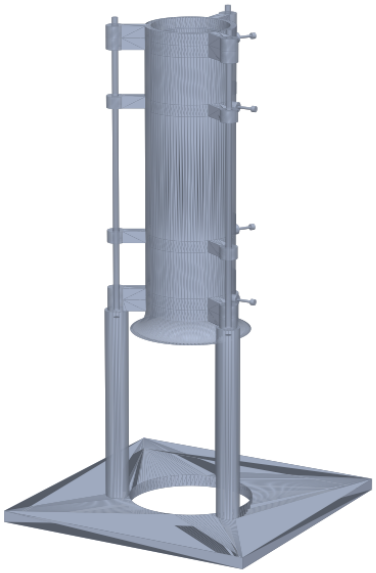
Ion Thruster - Type EDH

Design study of a compact multi-stage electrohydrodynamic (EHD) ionic-wind thruster

2-stage thruster on its test stand (exterior)



Author: roosterbrown14 · October 2026

Repository: github.com/roosterbrown14/Ion-Thruster-Type-EDH

Status. Modelling and design are complete. No prototype has been built and nothing has been measured yet. All numbers are simulation estimates; real hardware may differ by 2–3×. The modelling code, analysis and documentation were developed with the help of Claude (Anthropic), an AI assistant.

1. Summary

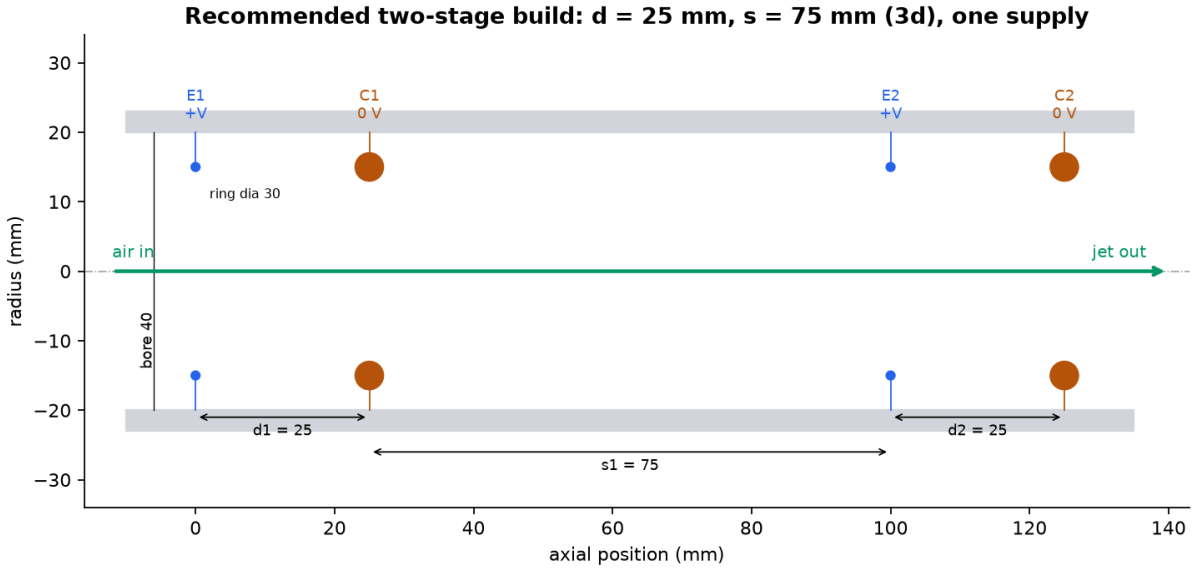
The thruster stacks ring-shaped emitters and collectors along a 40 mm tube. A positive corona at each emitter creates ions, which drift to the next collector and drag air with them, so the device produces thrust with no moving parts. The study asked how to choose the electrode gap, the stage spacing, the ring diameters, the pin layout, the materials and the number of stages.

The one rule behind everything. For every ion stream, the force on the air equals current × axial distance ÷ ion mobility, and the power equals current × voltage. Thrust per watt is therefore *(axial distance travelled) ÷ (mobility × voltage)*. Efficiency favours long gaps at low voltage; thrust favours high voltage. Ions that drift backwards subtract thrust.

Main findings

Finding	Detail
Stage spacing dominates multi-stage design	Ions from a later emitter drift back to the previous collector: 49 % of its current at spacing = gap, 16 % at 2×, 6 % at 3×. Keep spacing ≥ 3× the gap.
A guard ring removes back-flow	A smooth ring wired to the later emitter, 10–12 mm upstream, cuts back-flow to 0 %: +40 % thrust, and a 25 % shorter device. No extra supply needed.
Gap	Use the largest gap the supply can drive to about 85 % of sparkover (about 30–35 mm at 30 kV).
Stage count	More stages = more thrust at a fixed voltage, but lower thrust per watt. A third stage gave +20 % thrust for +53 % power.
Pins	Neighbouring needles shield each other: best at spacing ≈ pin length (12–16 pins on a 30 mm ring).
Collector	A thin, smooth aluminium tube aligned with the flow. Collector types differ little electrically.
Materials	Conductivity is irrelevant (losses about 10^{-8} of input). Stainless needles keep their tips.
Absolute performance	Recommended two-stage build: about 3 mN at about 1.3 W (about 2.3 mN/W) at 19.5 kV.

2. Concept and design



Source: illustrative hypothetical example
Section through the axis, to scale. Emitter: 0.1 mm-edge ring or pin ring; collector: 4 mm rounded ring (or 10 mm tube). Dimensions in mm. Supports must be insulators.

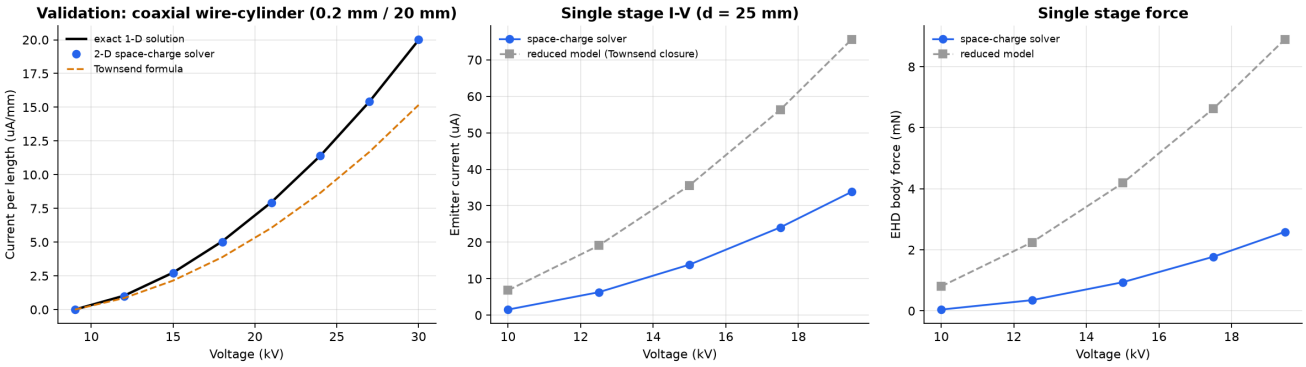
Recommended two-stage layout: 25 mm gaps, 75 mm stage spacing, 30 mm rings, 40 mm insulating bore.

Each stage carries 16 stainless needles on a brass ring (emitter) and a thin aluminium tube (collector). Both are held on three thin insulating spokes inside a 3-D-printed PETG/ASA housing, clamped by nylon tie rods. A rounded inlet reduces entry loss. One current-limited 30 kV supply feeds both emitters; collectors return to ground through separate current shunts, so back-flow can be measured directly.

3. Physics and models

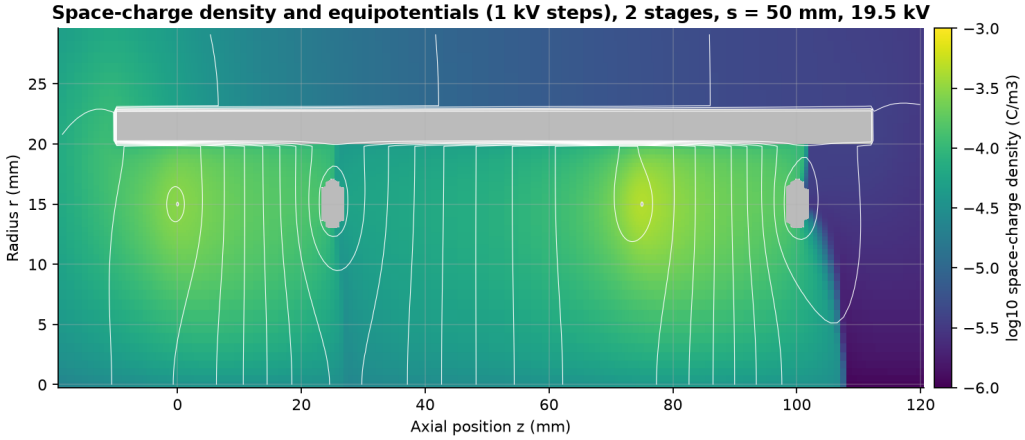
Two levels of model were used. The **reduced-order model** combines an exact axisymmetric electrostatic solver (charge simulation method), Peek's law for corona onset, Townsend's current law applied along traced ion paths, and a 1-D duct momentum balance. It is fast enough for parameter sweeps. A separate 3-D model handles discrete needles.

The **self-consistent space-charge solver** solves Poisson's equation and ion drift together on a graded axisymmetric grid, with Kaptsov's corona condition. It matches the exact solution for a wire in a cylinder to within 2 %, and its results do not depend on grid size or boundary placement. It showed the reduced model's absolute thrust is about 3× too high and its back-flow too low; the trends and rankings held.



Source: numerical simulation, self-consistent axisymmetric space-charge solver (verified vs exact coax; not validated by experiment)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, self-consistent space charge (Kaptsov injection). Insulating 40 mm bore modelled as a charged-up (zero-flux) wall.

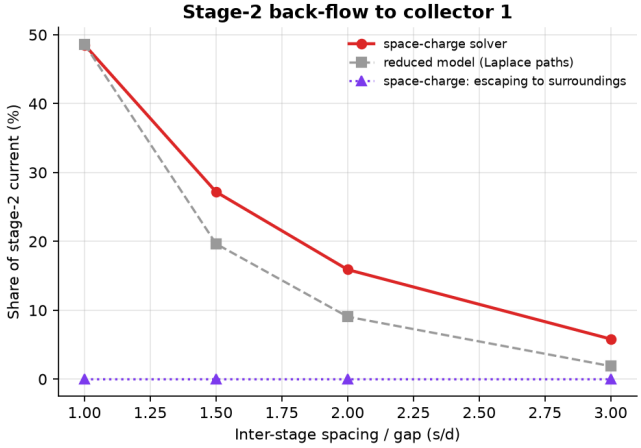
Left: solver validation against the exact coaxial solution. Middle and right: single-stage current and force, space-charge solver vs reduced model.

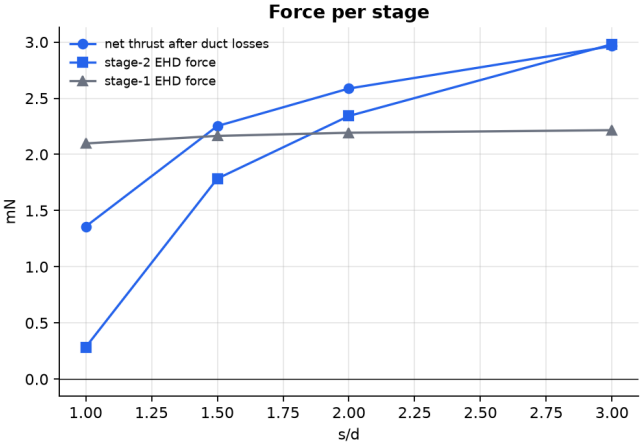


Source: numerical simulation, self-consistent axisymmetric space-charge solver (verified vs exact coax; not validated by experiment)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, self-consistent space charge (Kaptsov injection). Grey: electrodes and insulating bore wall.

Ion cloud (colour) and equipotentials in the two-stage device: back-flow is visible as charge leaking upstream from emitter 2.

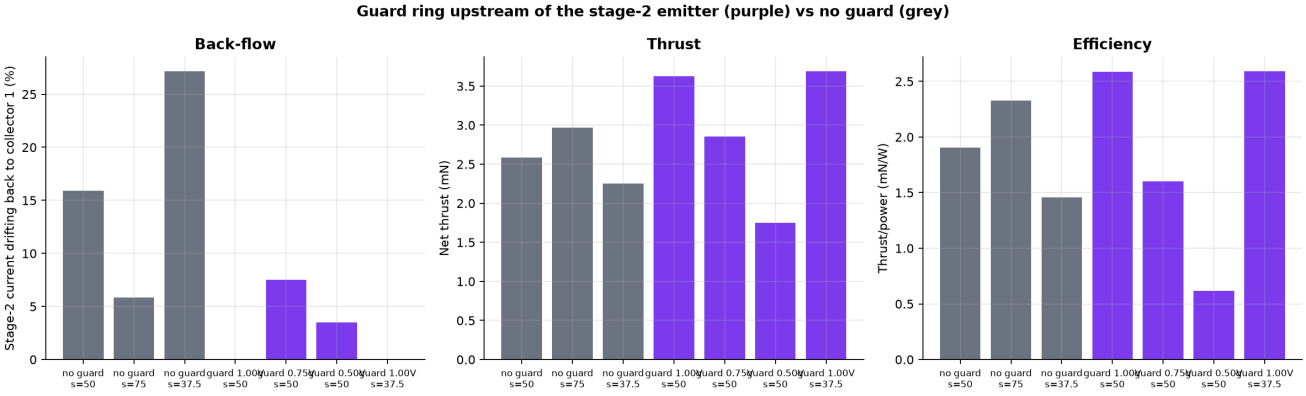
4. Key results





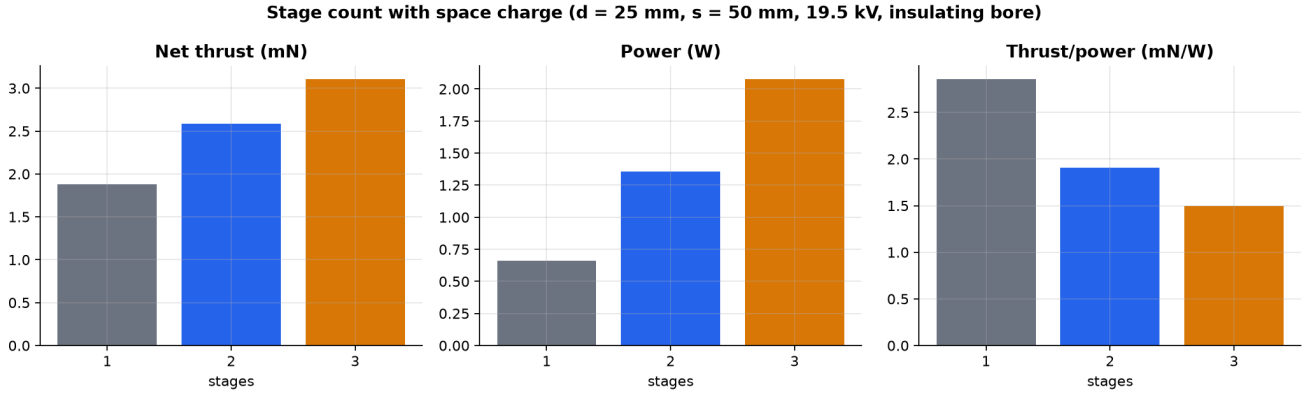
Source: numerical simulation, self-consistent axisymmetric space-charge solver (verified vs exact coax; not validated by experiment)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, self-consistent space charge (Kaptsov injection). Two stages, d = 25 mm, V = 19.5 kV.

Back-flow and per-stage force vs stage spacing (space-charge solver).



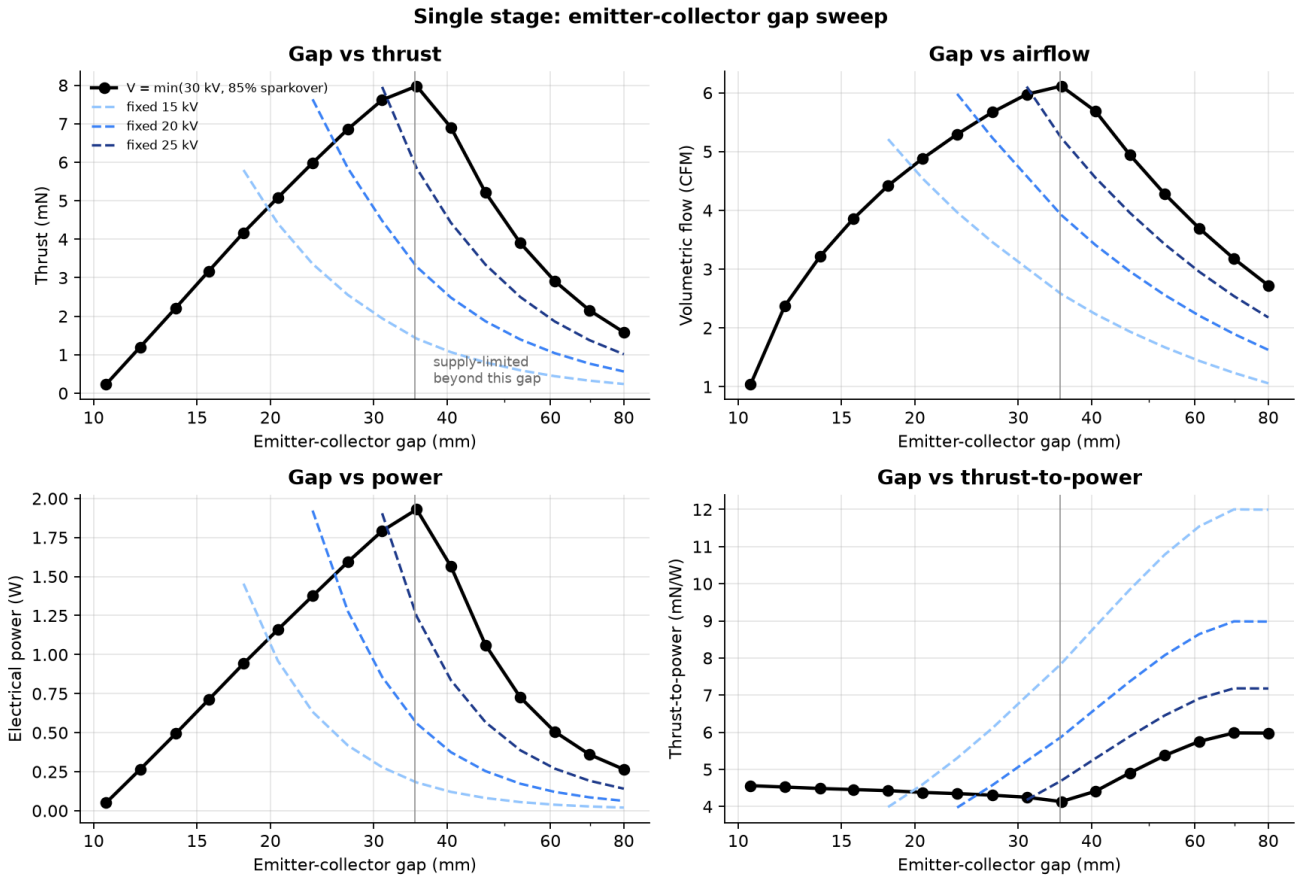
Source: numerical simulation, self-consistent axisymmetric space-charge solver (verified vs exact coax; not validated by experiment)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, self-consistent space charge. Two stages, d = 25 mm, 19.5 kV, insulating bore. Guard: 4 mm-section ring, 30 mm diameter. Guard 12 mm upstream of emitter 2 (10 mm at s = 37.5). Guard loss coefficient ASSUMED 0.3.

Guard ring at emitter voltage (purple) vs no guard (grey).



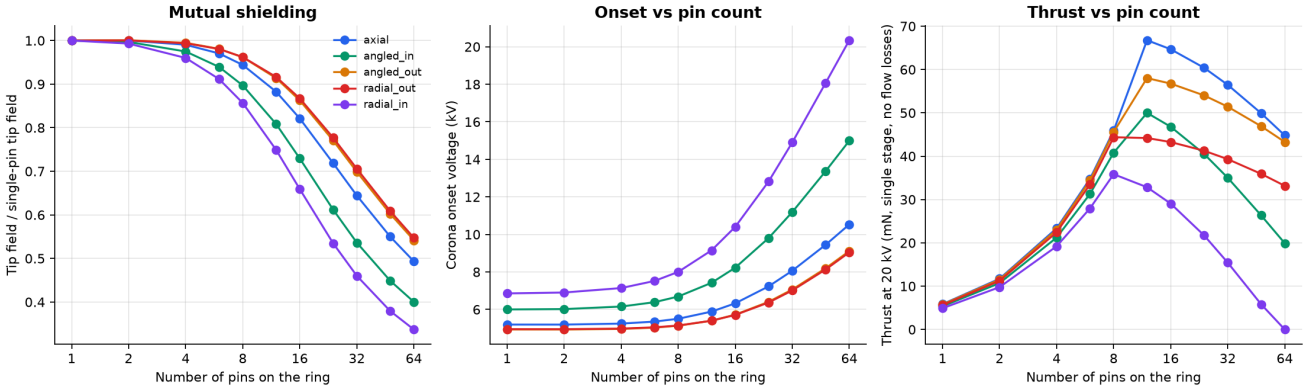
Source: numerical simulation, self-consistent axisymmetric space-charge solver (verified vs exact coax; not validated by experiment)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, self-consistent space charge (Kaptsov injection).

Stage count at the same voltage.



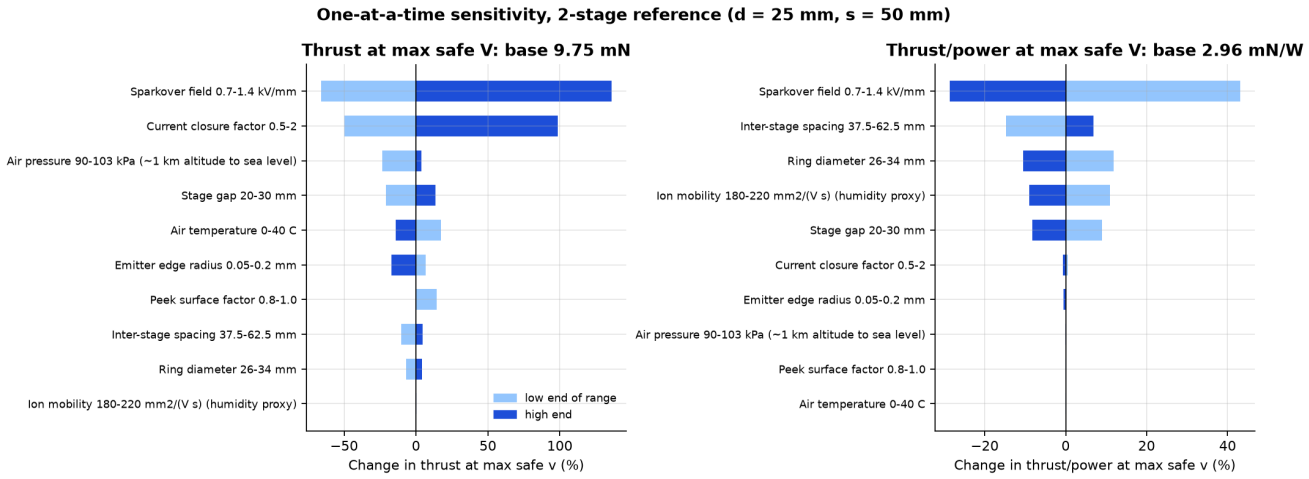
Source: numerical simulation, reduced-order CSM + Townsend model (unvalidated)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, no space charge in ion paths. Ring emitter, torus collector, both 30 mm dia.

Gap sweep (reduced model; trends reliable, absolute values about 2–3× high).



Source: numerical simulation, reduced-order CSM + Townsend model (unvalidated)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, no space charge in ion paths. Pins: 6 mm long, 0.05 mm tip radius, 30 mm ring, gap 25 mm (ring->collector), V=20 kV. Current = min(N x Sigmond pin current, ring Townsend ceiling): HEURISTIC; read trends, not absolute values.

Needle count, shielding and orientation (3-D model).



Source: numerical simulation, reduced-order CSM + Townsend model (unvalidated)
Assumptions: positive corona, mu = 200 mm2/(V s), sparkover 1.0 kV/mm mean field, Peek m=1, 40 mm bore, emitter edge r=0.1 mm, 20 C / 101.3 kPa, no space charge in ion paths. Humidity is represented only through the mobility and sparkover ranges; no humidity-specific model is used.

Sensitivity: sparkover field and current scale dominate the absolute numbers; gap and spacing dominate among design choices.

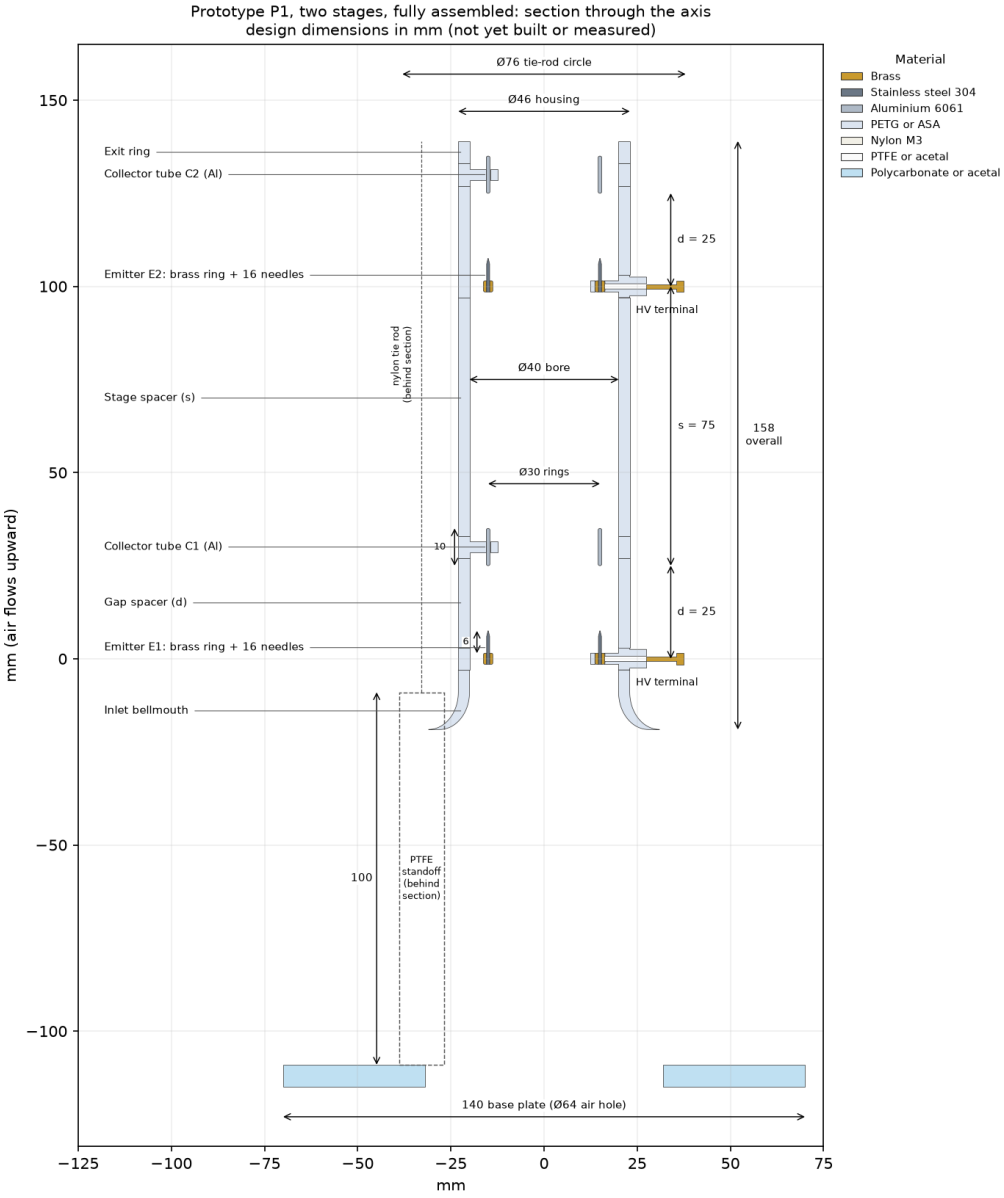
5. Recommendations

Factor	Best choice
Stage spacing	≥ 3× the gap, or a guard ring at emitter voltage upstream of each later emitter
Electrode gap	Largest the supply can drive to 85 % of sparkover (≈ 30–35 mm at 30 kV)
Number of stages	1 for efficiency, 2 for balance, 3 only for maximum thrust
Emitter	12–16 stainless needles (tip ≈ 0.05 mm) on a Ø30 ring; axial for thrust, radial for efficiency
Collector	Thin aluminium tube, same diameter as the emitter ring, edges rounded
Housing	Insulating tube (PETG/ASA). Never metal: it would arc.
Supply	One current-limited supply, unless per-stage gaps differ

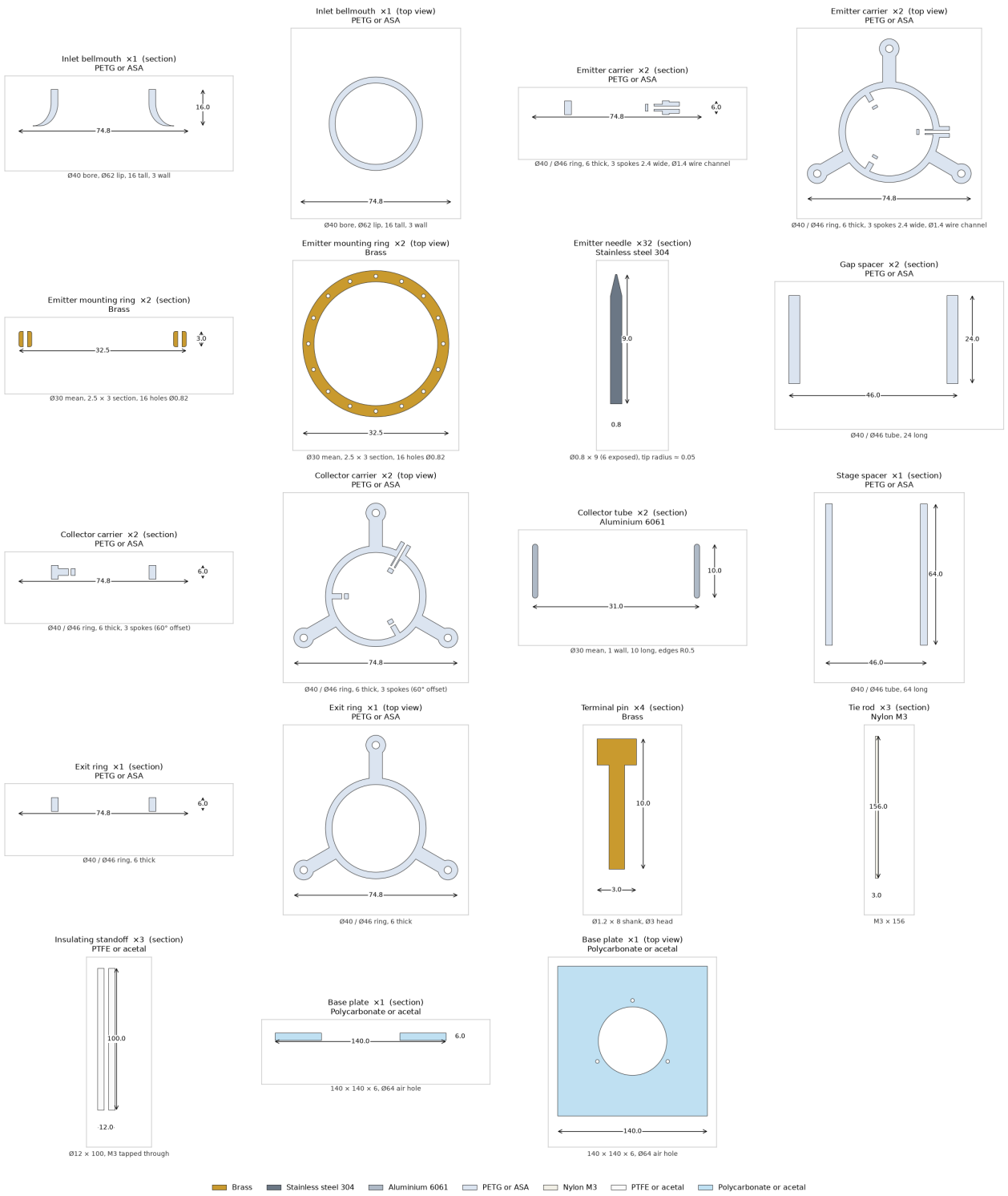
Configurations

Build	Configuration	Simulated performance
P1 (first build)	2 stages, d = 25 mm, s = 75 mm, 16 axial needles, Al tube collectors, 19.5 kV	≈ 3.0 mN, 1.3 W, 2.3 mN/W
P2 (after P1)	As P1 + guard ring wired to emitter 2, 10 mm upstream, s = 37.5 mm	≈ 3.7 mN, 1.4 W, 2.6 mN/W
Most efficient	1 stage, 30–35 mm gap, run just above onset	≈ 3–10 mN/W, under 1 mN

6. Prototype P1



Fully assembled P1 on its test stand, section through the axis, coloured by material (design dimensions in mm).



Every P1 part, separated, with material, quantity and dimensions.

Parts and materials

Part	Qty	Material	Key dimensions (mm)
Emitter needle	32	Stainless steel 304	Ø0.8 × 9, 6 exposed, tip ≈ 0.05
Emitter mounting ring	2	Brass	Ø30, 2.5 × 3 section, 16 holes
Collector tube	2	Aluminium 6061	Ø30, 1 wall, 10 long
Carriers, spacers, inlet, exit ring	9	PETG or ASA (printed)	Ø40 bore / Ø46 outside
Terminal pins	4	Brass	Ø1.2 × 8
Tie rods	3	Nylon	M3 × 156
Standoffs / base plate	3 / 1	PTFE or acetal / polycarbonate	Ø12 × 100 / 140 × 140 × 6

Thruster mass about 121 g. Indicative parts cost CAD $80–230; test equipment is normally available in a university lab.

7. Testing plan and safety

The models are unvalidated. The tests are ranked by how much they would change the conclusions:

#	Measurement	Why
1	Current to each collector separately, at several spacing/gap ratios	Back-flow decides stage count and spacing
2	Sparkover voltage vs clearance	Sets the optimum gap and the operating voltage
3	I–V curve per emitter type	Calibrates the current scale
4	Thrust on a precision balance (exhaust up, insulated stand)	Direct check of the main output
5	Needle count 8/12/16/24; collector type	Checks shielding and flow-loss assumptions

Safety. 10–30 kV DC can kill. Testing must be supervised, using a commercial current-limited supply inside an interlocked, grounded enclosure, with a grounding stick and no energised adjustments. Corona produces ozone and nitrogen oxides, so ventilation and an ozone monitor are required. Full procedure: docs/safety.md.

8. Limitations

- Nothing is validated against measurements yet.
- The space-charge solver treats needle rings as continuous rings.
- Sparkover uses a simple average-field limit (0.7–1.4 kV/mm), which dominates the absolute numbers.
- Electrode flow-loss coefficients are engineering estimates.
- Airflow is a 1-D duct model; no CFD.

9. Key references

Masuyama K., Barrett S.R.H. (2013) On the performance of electrohydrodynamic propulsion. Proc. R. Soc. A 469, 20120623. doi:10.1098/rspa.2012.0623
Gilmore C.K., Barrett S.R.H. (2015) Electrohydrodynamic thrust density using positive corona-induced ionic winds for in-atmosphere propulsion. Proc. R. Soc. A 471, 20140912. doi:10.1098/rspa.2014.0912
Xu H. et al. (2018) Flight of an aeroplane with solid-state propulsion. Nature 563, 532–535. doi:10.1038/s41586-018-0707-9
Gomez-Vega N. et al. (2023) Model of multistaged ducted thrusters for high-thrust-density electroaerodynamic propulsion. AIAA J. 61, 767–779. doi:10.2514/1.J061948
Sigmond R.S. (1982) J. Appl. Phys. 53, 891–898. doi:10.1063/1.330557 · Townsend J.S. (1914) Phil. Mag. 28, 83–90 · Peek F.W. (1929) Dielectric Phenomena in High-Voltage Engineering.
Full list (24 sources): references/references.bib